

STANDARD OPERATING PROCEDURE

Lockout/Tagout (LOTO) Procedure

Control of Hazardous Energy During Servicing and Maintenance

Document number	SOP-00087
Title	Lockout/Tagout (LOTO) Procedure
Revision	B.2 (revision B, iteration 2)
Lifecycle state	Released
Effective date	2026-02-02
Document owner	M. Chen, Environment, Health & Safety
Approved by	A. Brennan, Plant Manager (2026-01-22)
Applies to	Greenfield Plant – all production and facility equipment

1 Purpose

This procedure establishes the minimum requirements for the control of hazardous energy at Northwind Industrial. It ensures that machines and equipment are stopped, isolated from all energy sources and locked out before any employee performs servicing or maintenance where the unexpected energization, start-up or release of stored energy could cause injury.

- 2 Scope
- 2.1

This procedure applies to all employees and contractors who service, maintain, clean, adjust, set up or repair machines and equipment at the Greenfield Plant, and to employees who work in areas where such work is performed.
- 2.2

It applies to all forms of hazardous energy, including electrical, hydraulic, pneumatic, mechanical, thermal, chemical and gravitational energy.
- 2.3

Normal production operations are excluded unless a guard or safety device must be removed or bypassed, or a person must place any part of the body in the point of operation or another danger zone.
- 2.4

Work on cord-and-plug connected equipment is excluded when the plug is removed and remains under the exclusive control of the person performing the work.
- 2.5

Definitions:

Authorized employee:

A person trained and designated to lock out equipment in order to perform servicing or maintenance on it.

Affected employee:

A person who operates or uses the equipment being locked out, or who works in the area where the servicing is performed.

Energy control procedure (ECP):

The equipment-specific, written sequence for isolating and verifying all energy sources of a machine, posted at the machine.

Zero energy state:

The condition in which all energy sources are isolated and all stored or residual energy has been released or restrained.

3 References

Document	Title
29 CFR 1910.147	The control of hazardous energy (lockout/tagout)
ISO 14118	Safety of machinery – Prevention of unexpected start-up
EHS-002	Contractor Safety Management
EHS-014	Personal Protective Equipment Standard
F-087-01	Energy Control Procedure (ECP) template
F-087-02	Group Lockout Log
F-087-03	Annual Periodic Inspection Record
F-087-04	Lock Removal in Absence of the Authorized Employee

4 Responsibilities

Authorized employee: Applies personal locks and tags, verifies zero energy before starting work and removes only their own locks.

Affected employee: Recognizes lockout devices, never attempts to remove or bypass them and does not restart locked-out equipment.

Supervisor: Ensures that servicing is performed under this procedure, that ECPs are available at the machines and that only trained, authorized employees perform lockout.

Lockout coordinator: The single authorized employee with overall responsibility for a group lockout, including the group lock box and the Group Lockout Log F-087-02.

EHS Manager (document owner): Maintains this procedure, organizes training and the annual periodic inspections, and audits compliance.

Training

- 4.1** Authorized employees shall complete LOTO training, including a practical assessment on representative equipment, before performing lockout for the first time, and refresher training every 3 years.
- 4.2** Affected employees shall receive awareness training on the purpose and use of this procedure during induction.
- 4.3** Retraining is required whenever job assignments, equipment or ECPs change in a way that introduces a new hazard, or when a periodic inspection or an observation shows deviations from this procedure.
- 4.4** Contractors shall follow this procedure or an equivalent procedure approved by EHS in accordance with EHS-002.

5 Safety / PPE

WARNING Never rely on a machine's control circuit (emergency stop, stop button, interlock or PLC) for energy isolation. Only energy-isolating devices such as disconnect switches, circuit breakers and manual valves may be locked out.

- 5.1** PPE shall be selected in accordance with the ECP and EHS-014. As a minimum, safety glasses and safety shoes shall be worn.
- 5.2** The test for absence of voltage shall only be performed by electrically qualified persons wearing the arc-rated PPE stated on the equipment's arc-flash label.
- 5.3** Each authorized employee must use their own personal lock and key; keys shall not be shared.
- 5.4** Tags alone may not be used instead of locks unless the energy-isolating device cannot accept a lock and EHS has approved the tagout-only method in the ECP.

6 Tools & Materials

Item	Specification
Personal safety padlock	Red, keyed different, one key per lock, marked with the owner's name
Danger tag	"DANGER – DO NOT OPERATE", with name, department, date and phone
Lockout hasp	Steel, for up to 6 locks
Valve and breaker lockouts	Gate valve, ball valve and circuit breaker types
Group lock box	Holds the keys of all isolation locks of a group lockout
Voltage tester	Two-pole, CAT III 1000 V, proven on a known live source before and after use
Blocking devices	Mechanical blocks and safety pins rated for the load

7 Procedure

7.1 Preparation

- 7.1.1** Obtain the ECP for the equipment. If no ECP exists or the ECP does not match the equipment, stop and contact the Supervisor and EHS; work shall not begin until a valid ECP is available.
- 7.1.2** Identify every energy source and its isolation point from the ECP, including stored energy such as hydraulic accumulators, springs, capacitors, compressed air receivers and raised loads.
- 7.1.3** Notify all affected employees that the equipment will be shut down and locked out, and why.

7.2 Shutdown and isolation

- 7.2.1** Shut down the equipment using its normal stopping procedure.
- 7.2.2** Operate every energy-isolating device listed in the ECP, for example the main disconnect and the pneumatic and hydraulic isolation valves.
- 7.2.3** Apply a personal lock and a completed danger tag to each energy-isolating device. Where more than one person works on the equipment, use a hasp or a group lock box (see 7.5).

7.3 Stored energy

- 7.3.1** Hydraulic accumulators shall be bled down through the designated bleed valve until the gauge reads 0 bar. If the ECP lists no bleed valve, the accumulator shall be treated as charged and work shall not begin.
- 7.3.2** Pneumatic systems shall be vented to atmosphere and the lockable dump valve locked in the open position.
- 7.3.3** Suspended or raised parts, such as press rams, spindle heads and lifting tables, shall be lowered to a safe position or secured with mechanical blocks or safety pins rated for the load.
- 7.3.4** Capacitors and variable frequency drives shall be allowed to discharge for at least the time stated on the drive label (minimum 5 minutes) before the absence of voltage is tested.
- 7.3.5** If stored energy can re-accumulate, verification of isolation shall be continued until the work is completed.

7.4 Verification of zero energy

- 7.4.1** Confirm that all persons are clear of the equipment.
- 7.4.2** Attempt to start the equipment with the normal operating controls (try-out). The equipment must not start or move.
- 7.4.3** Test for absence of voltage at the load side of the isolation point using the live-dead-live method: test the voltage tester on a known live source, test all phases and each phase to earth, then test the voltage tester again on the known live source.
- 7.4.4** Check that all pressure gauges read 0 bar and that no parts can move under gravity.
- 7.4.5** Return all operating controls to the off or neutral position. Work may begin only after zero energy has been verified.

7.5 Group lockout

- 7.5.1** The lockout coordinator isolates the equipment, applies isolation locks to every energy-isolating device and verifies zero energy in accordance with 7.4.
- 7.5.2** The keys of the isolation locks are placed in the group lock box. Each member of the group applies a personal lock to the box before starting work and removes it when their own work is complete.
- 7.5.3** The lockout coordinator records all names on the Group Lockout Log F-087-02 and may remove the isolation locks only after all personal locks have been removed from the group lock box.
- 7.5.4** At shift change, the oncoming authorized employees shall apply their personal locks before the off-going employees remove theirs, so that lockout protection is continuous.

7.6 Release from lockout

- 7.6.1** Check that tools and blocking devices have been removed and that all guards and safety devices are reinstalled and functional.
- 7.6.2** Confirm that all persons are clear and notify the affected employees that the equipment will be re-energized.
- 7.6.3** Each lock shall be removed only by the authorized employee who applied it.
- 7.6.4** Remove the locks and tags, restore the energy sources in the order given in the ECP and perform a functional test of the equipment.

7.7 Removal of a lock in the absence of the authorized employee

- 7.7.1** If the authorized employee who applied a lock is not on site and cannot be reached, the lock may be removed only by the Maintenance Manager with an EHS representative present.
- 7.7.2** The removal shall be documented on F-087-04; the employee must be informed before resuming work.

8 Inspection & Acceptance Criteria

- 8.1** Each ECP shall be inspected at least annually by an authorized employee other than the persons using the procedure being inspected (annual periodic inspection).
- 8.2** The inspection shall include a review of the responsibilities of each authorized employee under the ECP and the observation of an actual or simulated lockout.
- 8.3** The inspection is acceptable only when all of the following are met:
- All energy sources and isolation points in the ECP match the installed equipment.
 - Every isolation point was locked with a personal lock and a completed tag.
 - Stored energy was released and zero energy was verified before work began.
 - The authorized employees could explain the ECP and their responsibilities.
- 8.4** Deficiencies shall be corrected within 30 days and the persons involved retrained. The inspector certifies the inspection on F-087-03 with the equipment, the date, the employees included and the inspector's name.

9 Records

- 9.1** ECPs are posted at each machine and reviewed whenever the equipment is modified.
- 9.2** Annual periodic inspection records (F-087-03), Group Lockout Logs (F-087-02) and lock removal records (F-087-04) shall be retained for 3 years.
- 9.3** Training records shall be retained for the duration of employment plus 3 years.

10 Revision History

Rev.	Date	Author	Description of change
A.1	2022-05-10	M. Chen	Initial release.
B.1	2025-06-30	M. Chen	Revision started: group lockout and shift change requirements added.
B.2	2026-01-22	M. Chen	Accumulator bleed-down verification and form F-087-04 added. Released.